

Disproof of a Claim About $\mathbb{E}[\hat{N}]$

Context (Sheet 02, Problem 2). In the capture–recapture simulation used in `sheet-02/code.ipynb`, the estimator of the population size is

$$\hat{N} = n + \hat{f}_0,$$

where f_k denotes the number of individuals caught exactly k times, $n = \sum_{k \geq 1} f_k$ is the number of distinct observed individuals, and

$$\hat{f}_0 = \begin{cases} \frac{f_1(f_1 - 1)}{2(f_2 + 1)}, & f_2 = 0, \\ \frac{f_1^2}{2f_2}, & f_2 > 0. \end{cases}$$

Claim to assess. The statement in question reads (rewritten with standard expectation notation):

“ $\mathbb{E}[\hat{N}] > N$, or if $\mathbb{E}[\hat{N}] = N$, then $\mathbb{E}[\hat{N} - N] > 0$ for $n < \infty$.”

Proposition 1 (The second clause is impossible). *For any integrable estimator $\hat{N}$ of a fixed (non-random) parameter N ,*

$$\mathbb{E}[\hat{N} - N] = \mathbb{E}[\hat{N}] - N.$$

In particular, if $\mathbb{E}[\hat{N}] = N$, then $\mathbb{E}[\hat{N} - N] = 0$.

Proof. Linearity of expectation yields

$$\mathbb{E}[\hat{N} - N] = \mathbb{E}[\hat{N}] - \mathbb{E}[N] = \mathbb{E}[\hat{N}] - N,$$

because N is a constant. The special case follows immediately. $\square$

Remark 1. *So the implication “if $\mathbb{E}[\hat{N}] = N$ then $\mathbb{E}[\hat{N} - N] > 0$ ” is false for all finite or infinite sample sizes; it contradicts basic properties of expectation.*

Proposition 2 (The strict inequality $\mathbb{E}(\hat{N}) > N$ is not generally true). *There is no general principle forcing $\mathbb{E}[\hat{N}] > N$ for estimators arising in capture models at finite sample size.*

Proof. Consider the natural estimator $\tilde{N} := n$ (the number of distinct observed individuals), which is a standard baseline in the same capture setup.

Assume a simple model consistent with the simulation spirit: there are N individuals, and across T capture occasions each individual is caught independently with per-occasion probability $p \in (0, 1)$. Then an individual is observed at least once with probability

$$q = 1 - (1 - p)^T \in (0, 1) \quad (T < \infty),$$

and therefore

$$n \sim \text{Binomial}(N, q) \quad \Rightarrow \quad \mathbb{E}[\tilde{N}] = \mathbb{E}[n] = Nq < N.$$

Hence even in a basic finite-sampling capture model, one can have $\mathbb{E}[\hat{N}] < N$ for a reasonable estimator. This disproves any unconditional claim that “the expected value must exceed N .” $\square$

Conclusion. The statement is *false as written*. The second clause is mathematically impossible, and the first clause is not a general truth (even within simple capture models for finite sampling).

How to proceed if you know the catch probability distribution (e.g. $p_i \sim U(0, a)$). If your goal is instead to study a *one-sided probability* like

$$\Pr(\hat{N} > N) > \Pr(\hat{N} \leq N) \iff \Pr(\hat{N} > N) > \frac{1}{2},$$

then you need distributional information about $\hat{N}$. In the capture setup of the notebook, a common route is:

Step 1: Specify the per-individual catch-count model. Fix the number of capture occasions $T < \infty$. For individual $i \in \{1, \dots, N\}$ let

$$K_i \mid p_i \sim \text{Binomial}(T, p_i),$$

where $p_i \in (0, 1)$ is that individual's per-occasion catch probability. Assume $(p_i)_{i=1}^N$ are i.i.d. from some distribution G (e.g. $U(0, a)$).

Step 2: Compute the mixed probabilities $q_k = \Pr(K_i = k)$. Unconditionally, the K_i are i.i.d. with the *mixture* pmf

$$q_k := \Pr(K_i = k) = \binom{T}{k} \mathbb{E}_{p \sim G} [p^k (1-p)^{T-k}], \quad k = 0, 1, \dots, T.$$

For $p \sim U(0, a)$ one gets an explicit integral:

$$q_k = \binom{T}{k} \frac{1}{a} \int_0^a p^k (1-p)^{T-k} dp = \binom{T}{k} \frac{B_a(k+1, T-k+1)}{a},$$

where $B_a(\alpha, \beta) = \int_0^a t^{\alpha-1} (1-t)^{\beta-1} dt$ is the (lower) incomplete beta function.

Step 3: Use the multinomial law for the frequency counts. Define $f_k := \sum_{i=1}^N \mathbf{1}\{K_i = k\}$. Since the K_i are i.i.d. with category probabilities $(q_0, \dots, q_T)$,

$$(f_0, f_1, \dots, f_T) \sim \text{Multinomial}(N; q_0, q_1, \dots, q_T).$$

In particular,

$$\mathbb{E}[f_k] = Nq_k, \quad \text{Var}(f_k) = Nq_k(1-q_k), \quad \text{Cov}(f_k, f_\ell) = -Nq_kq_\ell \quad (k \neq \ell).$$

Also $n = \sum_{k \geq 1} f_k = N - f_0$, hence $\mathbb{E}[n] = N(1 - q_0)$ and $\text{Var}(n) = Nq_0(1 - q_0)$.

Step 4: Approximate the distribution of $\hat{N}$ (delta method). Ignoring the $f_2 = 0$ fallback for the moment (see Step 5), the estimator used in the notebook is

$$\hat{N} = g(n, f_1, f_2) := n + \frac{f_1^2}{2f_2} \quad (f_2 > 0).$$

For large N , the multinomial central limit theorem implies

$$(n, f_1, f_2) \approx \mathcal{N}(m, \Sigma), \quad m = (N(1 - q_0), Nq_1, Nq_2),$$

with Σ determined by the multinomial covariances. The delta method then gives an approximate normal law

$$\hat{N} \approx \mathcal{N}(\mu_{\hat{N}}, \sigma_{\hat{N}}^2), \quad \mu_{\hat{N}} := g(m) = N \left(1 - q_0 + \frac{q_1^2}{2q_2} \right),$$

and

$$\sigma_{\hat{N}}^2 \approx \nabla g(m)^\top \Sigma \nabla g(m), \quad \nabla g(m) = \left(1, \frac{q_1}{q_2}, -\frac{q_1^2}{2q_2^2} \right).$$

Using $\text{Cov}(n, f_1) = Nq_0q_1$, $\text{Cov}(n, f_2) = Nq_0q_2$, $\text{Cov}(f_1, f_2) = -Nq_1q_2$, you can write $\sigma_{\hat{N}}^2$ explicitly as

$$\frac{\sigma_{\hat{N}}^2}{N} \approx q_0(1-q_0) + \left(\frac{q_1}{q_2} \right)^2 q_1(1-q_1) + \left(\frac{q_1^2}{2q_2^2} \right)^2 q_2(1-q_2) + 2 \left(\frac{q_1}{q_2} \right) q_0q_1 - 2 \left(\frac{q_1^2}{2q_2^2} \right) q_0q_2 - 2 \left(\frac{q_1}{q_2} \right) \left(\frac{q_1^2}{2q_2^2} \right) q_1q_2.$$

Step 5: Compute $\Pr(\hat{N} > N)$ and check if it exceeds 1/2. Under the normal approximation,

$$\Pr(\hat{N} > N) \approx 1 - \Phi \left(\frac{N - \mu_{\hat{N}}}{\sigma_{\hat{N}}} \right).$$

Therefore $\Pr(\hat{N} > N) > \frac{1}{2}$ holds approximately whenever $\mu_{\hat{N}} > N$, i.e.

$$1 - q_0 + \frac{q_1^2}{2q_2} > 1 \quad \Longleftrightarrow \quad \frac{q_1^2}{2q_2} > q_0,$$

and the inequality becomes more decisive the larger $(\mu_{\hat{N}} - N)/\sigma_{\hat{N}}$ is.

Step 6: Account for the $f_2 = 0$ fallback (if needed). In the multinomial model, $f_2 \sim \text{Binomial}(N, q_2)$ marginally, so

$$\Pr(f_2 = 0) = (1 - q_2)^N.$$

If Nq_2 is moderately large, this probability is negligible and the $f_2 > 0$ approximation dominates. Otherwise, you must include the fallback definition $\hat{f}_0 = \frac{f_1(f_1-1)}{2(f_2+1)}$ at $f_2 = 0$ and compute (analytically or by Monte Carlo) the mixture probability

$$\Pr(\hat{N} > N) = \Pr(\hat{N} > N \mid f_2 = 0) \Pr(f_2 = 0) + \Pr(\hat{N} > N \mid f_2 > 0) \Pr(f_2 > 0).$$